ECE 310 – Wireless Communications

M2 Technical Contribution

SOTA Synthesis, Research Gap, and Mathematical Formulation

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Abstract

This report presents M2 contribution for the project "Quantifying the Sum-Rate Cost of Reciprocity in Beyond-Diagonal RIS-Aided MU-MISO Systems." It synthesizes the main BD-RIS literature identified in the proposal, formulates the downlink MU-MISO system, defines SINR and sum-rate, and states the general-unitary and reciprocal symmetric-unitary constraints. A recent direct reciprocal/non-reciprocal comparison has been identified, so the broad claim that no prior comparison exists is not defensible. The proposed research gap is therefore framed as a working hypothesis requiring full-text verification.

Keywords: BD-RIS, MU-MISO, reciprocity, symmetric-unitary matrix, manifold optimization, sum-rate.

1 Background and Research Question

Conventional single-connected RIS uses a diagonal scattering matrix. Beyond-diagonal RIS (BD-RIS) permits non-diagonal scattering matrices and supports single-connected (CWSC), group-connected (CWGC), and fully-connected (CWFC) architectures. For a lossless network, the scattering matrix is unitary; reciprocity adds the symmetry condition $\Theta = \Theta^T$. The project measures the sum-rate difference between general unitary and reciprocal symmetric-unitary designs under matched channels and a shared fixed beamformer.

2 Critical Literature Synthesis

2.1 Li, Shen, and Clerckx (2023)

The base paper establishes a unified BD-RIS framework across single-, group-, and fully-connected architectures and RIS modes, and develops sum-rate optimization with joint transmit-precoder and scattering-matrix design. It provides the architecture model and general unitary benchmark. Its main framework does not isolate the added reciprocity constraint in the proposed matched fixed-beamformer comparison. DOI: https://doi.org/10.1109/TWC.2022.3210706.

2.2 Fidanovski et al. (2025 preprint)

The reciprocal BD-RIS work optimizes a symmetric scattering matrix. The preprint describes fractional programming, a symmetry penalty during manifold optimization, and final feasibility enforcement/projection. It provides the central reciprocal-design method; however, comparison only against reciprocal baselines does not itself quantify the rate cost relative to a general unitary design. Verify implementation details against the final publication. https://arxiv.org/abs/2509.20246.

2.3 Santamaria et al. (2026)

This work characterizes the symmetric-unitary manifold and develops Riemannian optimization methods that operate on the reciprocal feasible set. Exact manifold enforcement differs from penalty followed by projection. Its experiment-level evidence must be checked before claiming whether it quantifies the exact gap proposed here. https://arxiv.org/abs/2604.22469.

2.4 Critical competing paper: arXiv:2607.22170

A recent paper has been identified as direct competing work. The preliminary arXiv summary reports a reciprocal/non-reciprocal comparison across SC/GC/FC, with $N_t = N_r = 4$, $P_{\max} = 0$ dBm, 100 channel realizations, an RIS-size sweep, isotropic signalling rather than a transmit beamformer, and nearly coincident curves without numerical gap values. These details have not yet been checked against the full text. Therefore, "no one has compared reciprocal and non-reciprocal BD-RIS" is not a defensible claim. Full-text verification is required. https://arxiv.org/abs/2607.22170.

2.5 State-of-the-art comparison

Work	Focus	Method/scope	Relevance/limitation
Li et al. (2023)	General BD-RIS architecture and rate	Fractional programming/manifold optimization; SC/GC/FC; joint precoder and RIS	General benchmark; reciprocity cost not isolated under fixed-V protocol.
Fidanovski et al. (2025/2026)	Reciprocal BD-RIS	Symmetric-unitary design; penalty/manifold updates/projection in preprint	Reciprocal optimizer; compare directly with unitary benchmark.
Santamaria et al. (2026)	Symmetric-unitary geometry	Riemannian optimization	Relevant exact constraint handling; inspect experiments before gap claims.
Fidanovski et al. (arXiv:2607.22170)	Reciprocal vs non-reciprocal comparison, as preliminarily reported	Blockwise manifold method; isotropic signalling; RIS-size sweep	Critical competing work; verify details in full text.
Shen, Clerckx, and Murch (2022)	Scattering-parameter physical model	Network/scattering theory	Physical foundation; verify source details.
Nerini and Clerckx (2023)	Performance-complexity Pareto frontier	Architecture complexity/performance	Motivates complexity extension; does not alone establish the gap.
Zhou et al. (arXiv:2509.17804)	Generalized reciprocal architectures	Structure-oriented symmetric-unitary projection	Relevant reciprocal architecture work; no direct gap measurement identified in summary.
Wu and Clerckx (arXiv:2412.15965)	Universal BD-RIS optimization	pp-ADMM-style framework	Potential general baseline; reciprocity not the focus.
Yahya et al. (2024)	Passive MRT and interference nulling	Closed-form design	Explains use of off-diagonal terms; not sufficient alone to establish the gap.

3 Research Gap and Research Question

Working gap. The literature identified so far does not yet establish, under a matched fixed-MMSE-beamformer protocol, how the numerical reciprocity gap varies jointly with RIS size M, group configuration G, transmit power $P_{\max}$, and Rayleigh/Rician fading across CWGC and CWFC. This is a hypothesis to verify, not a proven first-ever claim.

Research question. Under identical channel realizations, a common fixed MMSE beamformer, equal power/noise settings, and matched architecture, how much sum-rate difference is observed between optimizing Θ over the unitary feasible set and over the symmetric-unitary reciprocal subset, and how does it vary with M, G, $P_{\max}$, and fading?

A near-zero gap is a valid result. The contribution is controlled quantification with paired gap estimates, confidence intervals, a CWSC zero-gap control, and feasibility/convergence checks.

4 System Model

Consider a downlink MU-MISO system with N BS transmit antennas, K single-antenna users, and an M-port BD-RIS. Let $H_{\mathrm{TX}} \in \mathbb{C}^{M\times N}$ be the BS-to-RIS channel, $H_{\mathrm{RX}} \in \mathbb{C}^{K\times M}$ the RIS-to-user channel, $\Theta \in \mathbb{C}^{M\times M}$ the scattering matrix, and $V = [v_1, \ldots, v_K] \in \mathbb{C}^{N\times K}$ the fixed beamformer. The data symbols satisfy $\mathbb{E}[ss^H] = I_K$ and receiver noise is $n \sim \mathcal{CN}(0, N_0 I_K)$.

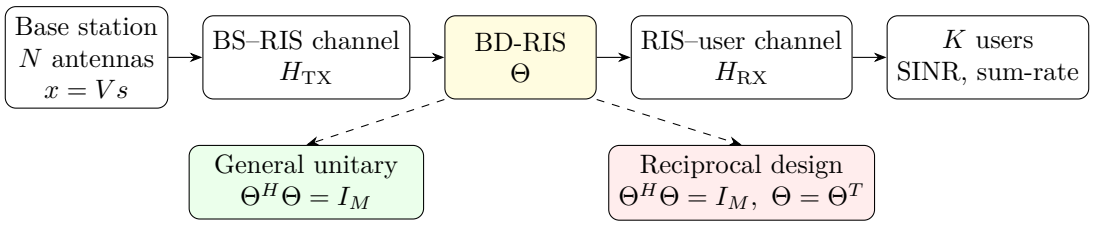


Figure 1: BD-RIS-aided MU-MISO downlink and the two scattering-matrix feasible sets.

Assuming the direct BS-to-user link is blocked in the baseline model, the transmit signal is

$$x = Vs, \qquad \|V\|_F^2 \le P_{\max}. \tag{1}$$

The received vector is

$$r = H_{\mathrm{RX}}\Theta H_{\mathrm{TX}} V s + n. \tag{2}$$

Define the effective channel

$$E = H_{\mathrm{RX}}\Theta H_{\mathrm{TX}} \in \mathbb{C}^{K\times N}, \tag{3}$$

and let e_k be row k of E. The stream coefficient at user k for stream i is

$$c_{ki} = e_k v_i. \tag{4}$$

Thus,

$$r_k = \sum_{i=1}^{K} c_{ki} s_i + n_k. \tag{5}$$

5 SINR and Sum-Rate

The desired power for user k is $|c_{kk}|^2$ and the multi-user interference is $\sum_{i\neq k} |c_{ki}|^2$. Hence,

$$\gamma_k = \frac{|c_{kk}|^2}{\sum\limits_{\substack{i=1\\ i\neq k}}^{K} |c_{ki}|^2 + N_0}, \quad k = 1, \ldots, K. \tag{6}$$

The spectral-efficiency sum-rate is

$$R_{\text{sum}}(\Theta; V) = \sum_{k=1}^{K} \log_2(1 + \gamma_k) \quad \text{bit/s/Hz}. \tag{7}$$

For fairness, compute one documented fixed V per channel realization, normalize it to the power budget, and pass the exact same V to both scattering-matrix optimizers.

6 Scattering-Matrix Constraints and Architectures

For a lossless passive multiport, power conservation means

$$\|\Theta a\|_2^2 = \|a\|_2^2, \quad \forall a \in \mathbb{C}^M, \tag{8}$$

which is equivalent to the unitary constraint

$$\Theta^H \Theta = I_M. \tag{9}$$

The general unitary set is

$$\mathcal{U}(M) = \{\Theta \in \mathbb{C}^{M \times M} : \Theta^H \Theta = I_M\}. \tag{10}$$

Reciprocity imposes

$$\Theta = \Theta^T. \tag{11}$$

The reciprocal feasible set is

$$\mathcal{U}_{\text{s}}(M) = \{\Theta \in \mathbb{C}^{M \times M} : \Theta^H \Theta = I_M,\ \Theta = \Theta^T\} \subseteq \mathcal{U}(M). \tag{12}$$

Here, "unconstrained" means unconstrained by reciprocity, not physically unconstrained in every respect.

For CWSC, Θ is diagonal:

$$\Theta_{\text{SC}} = \text{diag}(e^{j\phi_1}, \ldots, e^{j\phi_M}), \quad \phi_m \in . \tag{13}$$

For CWGC, it is block diagonal:

$$\Theta_{\text{GC}} = \text{blkdiag}(\Theta_1, \ldots, \Theta_G), \tag{14}$$

where $\Theta_g^H \Theta_g = I_{m_g}$ and, for reciprocal CWGC, $\Theta_g = \Theta_g^T$. For equal groups, $m_g = M/G$ and M/G must be an integer. For CWFC, Θ is a full unitary $M \times M$ matrix, additionally symmetric for the reciprocal design. Since diagonal unitary matrices are symmetric, CWSC has exactly zero theoretical reciprocity gap and is a mandatory control.

7 Reciprocity-Gap KPIs

For each matched channel realization ℓ, let $R_U^{(\ell)}$ and $R_S^{(\ell)}$ be the rates returned by the general and reciprocal optimizers. Define the absolute gap:

$$\Delta_{\text{abs}}^{(\ell)} = R_U^{(\ell)} - R_S^{(\ell)} \quad \text{[bit/s/Hz]}. \tag{15}$$

The relative gap is

$$\Delta_{\text{rel}}^{(\ell)} = 100 \frac{R_U^{(\ell)} - R_S^{(\ell)}}{R_U^{(\ell)}} \quad [\%], \quad R_U^{(\ell)} > 0. \tag{16}$$

For L matched channel realizations,

$$\overline{\Delta_{\text{abs}}} = \frac{1}{L} \sum_{\ell=1}^{L} \Delta_{\text{abs}}^{(\ell)}. \tag{17}$$

The sample standard deviation is

$$s_{\Delta_{\mathrm{abs}}} = \sqrt{\frac{1}{L-1}\sum_{\ell=1}^{L}\left(\Delta_{\mathrm{abs}}^{(\ell)} - \overline{\Delta_{\mathrm{abs}}}\right)^2}. \tag{18}$$

An approximate 95% confidence interval for the mean paired gap is

$$\overline{\Delta_{\mathrm{abs}}} \;\pm\; t_{0.975,L-1}\frac{s_{\Delta_{\mathrm{abs}}}}{\sqrt{L}}, \tag{19}$$

where $t_{0.975,L-1}$ is the 97.5th percentile of Student's t distribution with $L-1$ degrees of freedom. Report both rates, paired gaps, confidence intervals, negative measured-gap frequency, runtime, iteration counts, and feasibility errors. Local solvers may produce a negative measured gap even though the global unitary optimum cannot be worse; do not silently clip such results.

8 Channel Model

The proposal path-loss model is

$$\mathrm{PL}(d) = \zeta_0\left(\frac{d}{d_0}\right)^{-\epsilon}, \quad \zeta_0 = -30 \text{ dB},\ d_0 = 1 \text{ m},\ \epsilon = 2.2. \tag{20}$$

If $\mathrm{PL}(d)$ is a power gain, scale the complex channel amplitude by $\sqrt{\mathrm{PL}(d)}$. A Rician channel is

$$H = \sqrt{\mathrm{PL}(d)}\left(\sqrt{\frac{\kappa}{\kappa+1}}\overline{H}_{\mathrm{LoS}} + \sqrt{\frac{1}{\kappa+1}}\widetilde{H}_{\mathrm{NLoS}}\right), \tag{21}$$

where entries of $\widetilde{H}_{\mathrm{NLoS}}$ are i.i.d. $\mathcal{CN}(0,1)$ and κ is the linear Rician K-factor. Rayleigh fading corresponds to $\kappa = 0$:

$$H = \sqrt{\mathrm{PL}(d)}\,\widetilde{H}_{\mathrm{NLoS}}. \tag{22}$$

The LoS array response, geometry, user placement, direct-link assumption, Rician factor, and noise interpretation must be documented. Reproduce each paper using its own parameters; choose and state a separate common parameter set for the controlled comparison. Convert dBm to watts using

$$P[\mathrm{W}] = 10^{(P[\mathrm{dBm}]-30)/10}. \tag{23}$$

Thus, 20 dBm equals 0.1 W, and -80 dBm equals 10^{-11} W.

9 Fair Comparison and Validation Checklist

1. Use identical H_{TX}, H_{RX}, noise, power, architecture, fading realization, and other parameters for each pair.
2. Compute one fixed beamformer V per realization, normalize it once, and share it between both optimizers.
3. Check $\|V\|_F^2 \le P_{\max}$.
4. Check $\|\Theta^H\Theta - I_M\|_F$ for all solutions.
5. For reciprocal solutions, also check $\|\Theta - \Theta^T\|_F$ after final enforcement.
6. Validate SINR and sum-rate against a direct loop implementation; validate gradients using finite differences where applicable.

7. Save random seeds, parameters, convergence history, rates, and feasibility errors.

8. Run CWSC first: the theoretical reciprocity gap must be zero up to numerical tolerance.

9. Keep paper-specific baseline reproduction settings separate from common controlled-comparison settings.

10 Status and Outstanding Verification

SOTA synthesis, qualified research gap, system model, SINR and sum-rate, unitary and reciprocity constraints, KPI definitions, and validation protocol. Before final submission, read the full text of arXiv:2607.22170 and verify its experimental evidence; verify final metadata for the cited physical reciprocity and Pareto-frontier references; and integrate the shared MATLAB parameters and algorithm settings.

References

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